

Code:

```
#include <stdio.h>
int main() {
    int p, q;
    if (scanf("%d %d", &p, &q) != 2) {
        return 0;
    }
    char a[1005][1005];
    int top = 0, bottom = p - 1;
    int left = 0, right = q - 1;
    char ch = 'Y';
    while(top<=bottom && right>=left) {
        // Fill the top row
        for (int i = left; i <= right; i++) {
            a[top][i] = ch;
        }
        top++;
        // Fill the rightmost column
        for (int i = top; i <= bottom; i++) {
            a[i][right] = ch;
        }
        right--;
        // Fill the bottom row
        if (top <= bottom) {
            for (int i = right; i >= left; i--) {
                a[bottom][i] = ch;
            }
            bottom--;
        }
        // Fill the leftmost column
        if (left <= right) {
            for (int i = bottom; i >= top; i--) {
                a[i][left] = ch;
            }
            left++;
        }
        // Alternate character for the next inner boundary
        ch = (ch == 'Y') ? 'O' : 'Y';
    }
    // Print the matrix
    for (int i = 0; i < p; i++) {
        for (int j = 0; j < q; j++) {
            printf("%c", a[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Input:

6 7

Output:

```
YYYYYYY
Y00000Y
Y0Y0Y0Y
Y0Y0Y0Y
Y00000Y
YYYYYYY
```